

# **Bangladesh University of Business & Technology (BUBT)**



## **LAB Report**

Course Code : CSE 210  
Course Title : Operating Systems Lab  
Experiment No. : 04

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Section: 01  
Program: B.Sc. Engg. in CSE

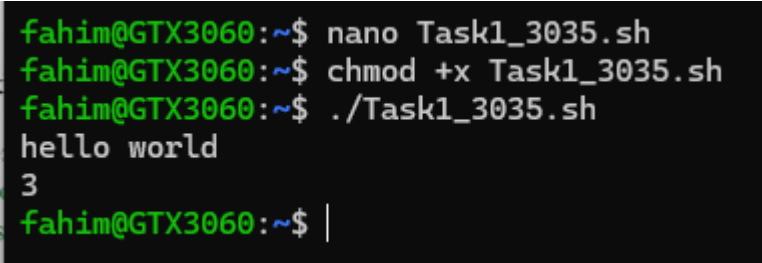
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\_\_\_\_\_  
**Signature of Teacher**

## 1. Write a shell script to count the number of vowels in a given string.

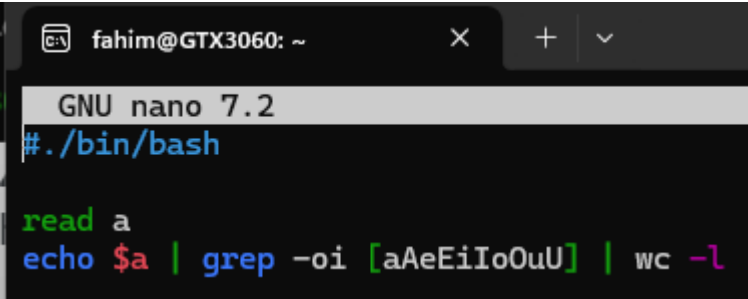
```
nano Task1_3035.sh
chmod +x Task1_3035.sh
./Task1_3035.sh
```



```
fahim@GTX3060:~$ nano Task1_3035.sh
fahim@GTX3060:~$ chmod +x Task1_3035.sh
fahim@GTX3060:~$ ./Task1_3035.sh
hello world
3
fahim@GTX3060:~$ |
```

### Inside Task1\_3035.sh:

```
#!/bin/bash
read a
echo $a | grep -oi [aAeEiloOuU] | wc -l
```

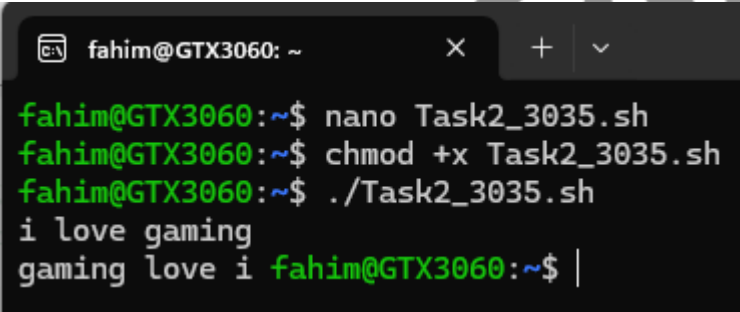


```
fahim@GTX3060: ~
GNU nano 7.2
#!/bin/bash

read a
echo $a | grep -oi [aAeEiIoOuU] | wc -l
```

## 2. Write a shell script that takes a sentence as input and outputs the sentence with the words reversed.

```
nano Task2_3035.sh
chmod +x Task2_3035.sh
./Task2_3035.sh
```



```
fahim@GTX3060:~$ nano Task2_3035.sh
fahim@GTX3060:~$ chmod +x Task2_3035.sh
fahim@GTX3060:~$ ./Task2_3035.sh
i love gaming
gaming love i fahim@GTX3060:~$ |
```

### Inside Task2\_3035.sh:

```
#!/bin/bash
read a
echo $a | awk '{for(i=NF; i>0; i--) printf "%s ", $i}'
```

```
fahim@GTX3060: ~  
GNU nano 7.2  
#./bin/bash  
read a  
echo $a | grep -oi [aAeEiIoOuU] | wc -l
```

3. Write a shell script to print numbers from 1 to nnn. For multiples of 3, print "Fizz", for multiples of 5 print "Buzz", and for numbers which are multiples of both 3 & 5, print "FizzBuzz".

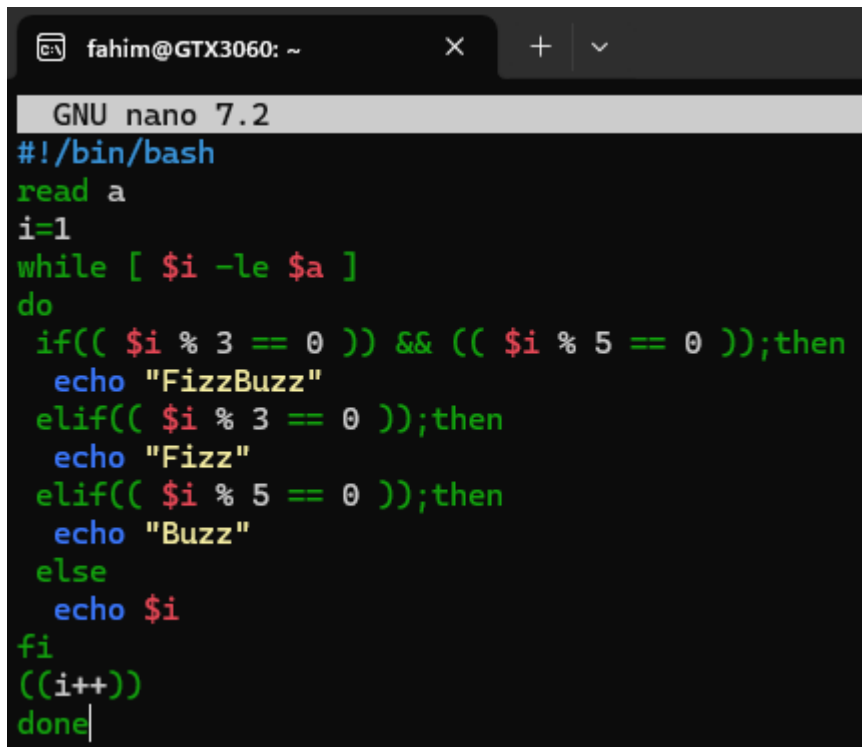
```
nano Task3_3035.sh  
chmod +x Task3_3035.sh  
./Task3_3035.sh
```

```
fahim@GTX3060: ~  
fahim@GTX3060:~$ nano Task3_3035.sh  
fahim@GTX3060:~$ chmod +x Task3_3035.sh  
fahim@GTX3060:~$ ./Task3_3035.sh  
15  
1  
2  
Fizz  
4  
Buzz  
Fizz  
7  
8  
Fizz  
Buzz  
11  
Fizz  
13  
14  
FizzBuzz  
fahim@GTX3060:~$ |
```

Inside Task3\_3035.sh:

```
#!/bin/bash  
read a  
i=1  
while [ $i -le $a ]  
do
```

```
if(( $i % 3 == 0 )) && (( $i % 5 == 0 ));then
    echo "FizzBuzz"
elif(( $i % 3 == 0 ));then
    echo "Fizz"
elif(( $i % 5 == 0 ));then
    echo "Buzz"
else
    echo $i
fi
((i++))
done
```

A screenshot of a terminal window with a dark background. The window title is 'fahim@GTX3060: ~'. Inside the terminal, the GNU nano 7.2 editor is open, showing a shell script. The script starts with a shebang '#!/bin/bash', followed by 'read a' to get input, 'i=1' to initialize a counter, and a 'while' loop 'while [ \$i -le \$a ]'. Inside the loop is an 'if' statement with three 'elif' branches: 'if(( \$i % 3 == 0 )) && (( \$i % 5 == 0 ));then' (FizzBuzz), 'elif(( \$i % 3 == 0 ));then' (Fizz), and 'elif(( \$i % 5 == 0 ));then' (Buzz). An 'else' branch prints the number '\$i'. The loop ends with 'fi', an increment '((i++))', and 'done'. The cursor is at the end of the 'done' line.

```
GNU nano 7.2
#!/bin/bash
read a
i=1
while [ $i -le $a ]
do
    if(( $i % 3 == 0 )) && (( $i % 5 == 0 ));then
        echo "FizzBuzz"
    elif(( $i % 3 == 0 ));then
        echo "Fizz"
    elif(( $i % 5 == 0 ));then
        echo "Buzz"
    else
        echo $i
    fi
    ((i++))
done
```

**4. Write a shell script that takes a list of integers as input and removes duplicates.**

```
nano Task4_3035.sh
chmod +x Task4_3035.sh
./Task4_3035.sh
```

```
fahim@GTX3060: ~  
fahim@GTX3060:~$ nano Task4_3035.sh  
fahim@GTX3060:~$ chmod +x Task4_3035.sh  
fahim@GTX3060:~$ ./Task4_3035.sh  
1 2 3 2 4 5 1  
1 2 3 4 5  
fahim@GTX3060:~$ |
```

### Inside Task4\_3035.sh:

```
#!/bin/bash  
read -a a  
x=""  
i=0  
while [ $i -lt ${#a[@]} ]; do  
    num="${a[$i]}"  
    if [[ ! " $b " =~ " $num " ]]; then  
        b+="$num "  
    fi  
    ((i++))  
done  
echo "$b"
```

```
GNU nano 7.2  
#!/bin/bash  
read -a a  
x=""  
i=0  
while [ $i -lt ${#a[@]} ]; do  
    num="${a[$i]}"  
    if [[ ! " $b " =~ " $num " ]]; then  
        b+="$num "  
    fi  
    ((i++))  
done  
echo "$b"
```

5. Write a shell script that takes two strings as input and prints the common characters between them.

```
nano Task5_3035.sh  
chmod +x Task5_3035.sh  
./Task5_3035.sh
```

```
fahim@GTX3060: ~  
fahim@GTX3060:~$ nano Task5_3035.sh  
fahim@GTX3060:~$ chmod +x Task5_3035.sh  
fahim@GTX3060:~$ ./Task5_3035.sh  
hello world  
welcome  
ellowol  
fahim@GTX3060:~$ |
```

Inside Task5\_3035.sh:

```
#!/bin/bash  
read s1  
read s2  
i=0  
result=""  
while [ $i -lt ${#s1} ]; do  
    ch="${s1:i:1}"  
    if [[ "$s2" == *"$ch"* ]]; then  
        result="$result$ch"  
    fi  
    i=$((i+1))  
done  
echo "$result"
```

```
GNU nano 7.2  
#!/bin/bash  
read s1  
read s2  
i=0  
result=""  
while [ $i -lt ${#s1} ]; do  
    ch="${s1:i:1}"  
    if [[ "$s2" == *"$ch"* ]]; then  
        result="$result$ch"  
    fi  
    i=$((i+1))  
done  
echo "$result"
```